Food Waste Reduction Mobile Application: Real-Time Surplus Redistribution Platform

Introduction

9.7M
Tons Wasted/Year

Bogotá
Colombia

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Cafeterias & Restaurants

In Colombia, 9.7 million tons of food are wasted annually. Specifically in Bogotá, cafeterias and local restaurants generate daily surpluses suitable for consumption, but the lack of rapid coordination and the short shelf life lead to their loss, exacerbating environmental damage, food insecurity, and social inequality. Platforms such as OLIO and EatCloud have demonstrated the potential of mobile applications; however, none are optimized for accessible environments with a strict 3 km walking radius, same-day operation, and mandatory priority for verified foundations.

For this reason, this project proposes a mobile platform specifically designed for the area surrounding the Faculty of Engineering at the Universidad Distrital Francisco José de Caldas, aimed at addressing these limitations through precise geolocation, proximity-based matching with priority for charities, and real-time notifications.

Goal

Research Question

How can a mobile platform, constrained to a 3 km walking-access radius, enable efficient real-time redistribution of food surpluses through geolocation and priority-based matching, in order to reduce food waste and improve equitable access?

Expected Product

A functional mobile platform prototype that enables real-time redistribution of food surpluses within a 3 km walking-access radius, integrating geolocation services, a proximity-based matching algorithm with priority rules for verified organizations, and a real-time notification system. The product is expected to demonstrate measurable improvements in surplus recovery rate, matching efficiency, and successful pickup rates under concurrent user scenarios.